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Outdoor air hood assembly with an inlet hood

Abstract

A hood assembly for a heating, ventilation, and air conditioning (HVAC) unit includes a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to rotatably couple to a housing of the HVAC unit. The hood assembly further includes a filter frame rotatably coupled to the top panel, wherein the filter frame is configured to support at least one filter. The hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel portion are configured to translate relative to one another during transition of the hood assembly between the collapsed configuration and the deployed configuration, and the top panel is configured to contain the filter frame within the housing in the collapsed configuration.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/235,726, entitled “AN OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD,” filed Apr. 20, 2021, which claims priority from and the benefit of India Provisional Application Serial No. 202011016873, entitled “AN OUTDOOR AIR HOOD ASSEMBLY WITH AN INLET HOOD,” filed Apr. 20, 2020, each of which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure and are described below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be noted that these statements are to be read in this light, and not as admissions of prior art.

(2) Heating, ventilation, and air conditioning (HVAC) systems are utilized to control environmental properties, such as temperature and humidity, for occupants of residential, commercial, and industrial environments. The HVAC systems may control the environmental properties through control of an air flow delivered to the environment. In some cases, the HVAC systems include an intake hood that is configured to block water, dust, debris, and other contaminants from entering the HVAC system. For instance, an outdoor HVAC unit may draw air into the unit from the intake hood, and the air may ultimately flow across a heat exchanger to exchange thermal energy with a working fluid (e.g., refrigerant) flowing through the heat exchanger. Existing intake hoods are generally shipped separately from a housing of the HVAC unit and are therefore assembled upon delivery, which increases transportation, delivery, and installation costs and may be time-consuming. Accordingly, it is now recognized that improved assembly and transportation management for HVAC units having intake hoods is desired.

SUMMARY

(3) A summary of certain embodiments disclosed herein is set forth below. It should be noted that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(4) In an embodiment, a hood assembly for a heating, ventilation, and air conditioning (HVAC) unit comprises a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to rotatably couple to a housing of the HVAC unit. The HVAC unit further comprises a filter frame rotatably coupled to the top panel, wherein the filter frame is configured to support at least one filter. The hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel portion are configured to translate relative to one another during transition of the hood assembly between the collapsed configuration and the deployed configuration, and the top panel is configured to contain the filter frame within the housing in the collapsed configuration.

(5) In another embodiment, a heating, ventilation, and air conditioning (HVAC) unit comprises a housing and an intake hood assembly coupled to the housing and configured to transition between a collapsed configuration and a deployed configuration. The HVAC unit further comprises a top cover of the intake hood rotatably coupled to the housing and comprising a first panel and a second panel, wherein the second panel is configured to translate relative to the first panel during transition of the intake hood assembly between the collapsed configuration and the deployed configuration. The HVAC unit further comprises a filter frame rotatably coupled to the top cover and configured to support at least one filter, wherein the top cover and the filter frame are configured to rotate relative to one another and relative to the housing during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

(6) In another embodiment, an intake hood assembly for a heating, ventilation, and air conditioning (HVAC) unit includes a top panel comprising a first panel portion and a second panel portion adjustably coupled to one another, wherein the top panel is configured to couple to a housing of the HVAC unit. The intake hood assembly further includes a filter frame coupled to the top panel, wherein the filter frame comprises a first sub-frame configured to support a first filter and a second sub-frame configured to support a second filter. The intake hood assembly is adjustable between a collapsed configuration and a deployed configuration, the first panel portion and the second panel

portion are configured to translate relative to one another during transition of the intake hood assembly between the collapsed configuration and the deployed configuration, and the first sub-frame and the second sub-frame are configured to rotate relative to one another during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:
- (2) FIG. 1 is a perspective view of a building having an embodiment of a heating, ventilation, and air conditioning (HVAC) system for environmental management that may employ one or more HVAC units, in accordance with an aspect of the present disclosure;
- (3) FIG. 2 is a perspective view of an embodiment of a packaged HVAC unit that may be used in the HVAC system of FIG. 1, in accordance with an aspect of the present disclosure;
- (4) FIG. 3 is a cutaway perspective view of an embodiment of a residential, split HVAC system, in accordance with an aspect of the present disclosure;
- (5) FIG. 4 is a schematic illustration of an embodiment of a vapor compression system that can be used in any of the systems of FIGS. 1-3, in accordance with an aspect of the present disclosure;
- (6) FIG. 5 is a perspective view of an embodiment of a portion of an HVAC unit having an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;
- (7) FIG. 6 is a perspective view of an embodiment of an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;
- (8) FIG. 7A is a cross-sectional side view of an embodiment of a top panel retainer of an air hood assembly in a deployed configuration, in accordance with an aspect of the present disclosure;
- (9) FIG. 7B is a cross-sectional side view of an embodiment of a top panel retainer of an air hood assembly in a collapsed configuration, in accordance with an aspect of the present disclosure;
- (10) FIG. 8A is a schematic side view illustrating transition of an embodiment of an air hood assembly between a deployed configuration and a collapsed configuration, in accordance with an aspect of the present disclosure;
- (11) FIG. 8B is a perspective view of an embodiment of an air hood assembly, illustrating transition between a deployed configuration and a collapsed configuration, in accordance with an aspect of the present disclosure;
- (12) FIG. 9 is a perspective view of an embodiment of an air hood assembly, in accordance with an aspect of the present disclosure;
- (13) FIG. 10A is an expanded perspective view of an embodiment of an air hood assembly, illustrating panel portions of the air hood assembly, in accordance with an aspect of the present disclosure;
- (14) FIG. 10B is an expanded perspective view of an embodiment of an air hood assembly, illustrating coupled panel portions of the air hood assembly, in accordance with an aspect of the present disclosure;
- (15) FIG. 11A is a perspective view of an embodiment of a connection between filter frames of an air hood assembly, in accordance with an aspect of the present disclosure;
- (16) FIG. 11B is a perspective view of an embodiment of a connection between a filter frame of an air hood assembly and a housing frame, in accordance with an aspect of the present disclosure;
- (17) FIG. 12 is a perspective view of an embodiment of an air hood assembly in a collapsed configuration, in accordance with an aspect of the present disclosure; and
- (18) FIG. 13 is a schematic of an embodiment of a filter assembly of an air hood assembly, in accordance with an aspect of the present disclosure.

DETAILED DESCRIPTION

(19) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be noted that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(20) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be noted that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features.

(21) The present disclosure is directed to an improved hood assembly (e.g., air hood assembly, air intake hood assembly) configured to be disposed over an air intake (e.g., air inlet) of a heating, ventilation, and air conditioning (HVAC) unit. As discussed above, existing air hoods may be removed and/or otherwise disassembled to an HVAC unit when the HVAC unit is to be transported from one location to another. As such, the HVAC unit and the air hood are shipped to a new location separately from one another, which may increase transportation costs. The air hood must also be assembled, or otherwise attached, to the HVAC unit when the separate components reach the final destination, which may be time-consuming and costly.

(22) Accordingly, embodiments of the present disclosure are directed to a collapsible air hood assembly that is configured to transition between a collapsed configuration and a deployed configuration. In some embodiments, the air hood assembly may include a top panel having a first panel and a second panel and having a pair of side panels removably coupled to the top panel. When the air hood assembly is in the deployed configuration, the air hood assembly is configured to block water and other elements, such as dust, debris, and/or dirt particles present in an environment surrounding the HVAC unit, from entering the HVAC unit via an air intake, while also enabling air to flow from the environment surrounding the HVAC unit into the HVAC unit via the air intake. When the air hood assembly is in the collapsed configuration, the pair of side panels may be removed such that the top panel of the air hood assembly may transition and rest generally flush with a side of the HVAC unit. As such, the air hood assembly may be transported with the HVAC unit in an assembled or installed configuration without substantially increasing the size (e.g., footprint) of the HVAC unit, thereby reducing the time and costs associated with transportation, delivery, and assembly of the HVAC unit.

(23) Turning now to the drawings, FIG. 1 illustrates a heating, ventilation, and air conditioning (HVAC) system for building environmental management that may employ one or more HVAC units. As used herein, an HVAC system includes any number of components configured to enable regulation of parameters related to climate characteristics, such as temperature, humidity, air flow, pressure, air quality, and so forth. For example, an “HVAC system” as used herein is defined as conventionally understood and as further described herein. Components or parts of an “HVAC system” may include, but are not limited to, all, some of, or individual parts such as a heat exchanger, a heater, an air flow control device, such as a fan, a sensor configured to detect a climate characteristic or operating parameter, a filter, a control device configured to regulate operation of an HVAC system component, a component configured to enable regulation of climate characteristics, or a combination thereof. An “HVAC system” is a system configured to provide

such functions as heating, cooling, ventilation, dehumidification, pressurization, refrigeration, filtration, or any combination thereof. The embodiments described herein may be utilized in a variety of applications to control climate characteristics, such as residential, commercial, industrial, transportation, or other applications where climate control is desired.

(24) In the illustrated embodiment, a building **10** is air conditioned by a system that includes an HVAC unit **12**. The building **10** may be a commercial structure or a residential structure. As shown, the HVAC unit **12** is disposed on the roof of the building **10**; however, the HVAC unit **12** may be located in other equipment rooms or areas adjacent the building **10**. The HVAC unit **12** may be a single package unit containing other equipment, such as a blower, integrated air handler, and/or auxiliary heating unit. In other embodiments, the HVAC unit **12** may be part of a split HVAC system, such as the system shown in FIG. 3, which includes an outdoor HVAC unit **58** and an indoor HVAC unit **56**.

(25) The HVAC unit **12** is an air cooled device that implements a refrigeration cycle to provide conditioned air to the building **10**. Specifically, the HVAC unit **12** may include one or more heat exchangers across which an air flow is passed to condition the air flow before the air flow is supplied to the building. In the illustrated embodiment, the HVAC unit **12** is a rooftop unit (RTU) that conditions a supply air stream, such as environmental air and/or a return air flow from the building **10**. After the HVAC unit **12** conditions the air, the air is supplied to the building **10** via ductwork **14** extending throughout the building **10** from the HVAC unit **12**. For example, the ductwork **14** may extend to various individual floors or one or more zones (**101**, **102**, **103**) of the building **10** and each zone may further comprise one or more outdoor air hoods equipped with filters. In certain embodiments, the HVAC unit **12** may be a heat pump that provides both heating and cooling to the building with one refrigeration circuit configured to operate in different modes. In other embodiments, the HVAC unit **12** may include one or more refrigeration circuits for cooling an air stream and a furnace for heating the air stream.

(26) A control device **16**, one type of which may be a thermostat, may be used to designate the temperature of the conditioned air. The control device **16** also may be used to control the flow of air through the ductwork **14**. For example, the control device **16** may be used to regulate operation of one or more components of the HVAC unit **12** or other components, such as dampers and fans, within the building **10** that may control flow of air through and/or from the ductwork **14**. In some embodiments, other devices may be included in the system, such as pressure and/or temperature transducers or switches that sense the temperatures and pressures of the supply air, return air, and so forth. Moreover, the control device **16** may include computer systems that are integrated with or separate from other building control or monitoring systems, and even systems that are remote from the building **10**.

(27) FIG. 2 is a perspective view of an embodiment of the HVAC unit **12**. In the illustrated embodiment, the HVAC unit **12** is a single package unit that may include one or more independent refrigeration circuits and components that are tested, charged, wired, piped, and ready for installation. The HVAC unit **12** may provide a variety of heating and/or cooling functions, such as cooling only, heating only, cooling with electric heat, cooling with dehumidification, cooling with gas heat, or cooling with a heat pump. As described above, the HVAC unit **12** may directly cool and/or heat an air stream provided to the building **10** to condition a space in the building **10**.

(28) As shown in the illustrated embodiment of FIG. 2, a cabinet **24** encloses the HVAC unit **12** and provides structural support and protection to the internal components from environmental and other contaminants. In some embodiments, the cabinet **24** may be constructed of galvanized steel and insulated with aluminum foil faced insulation. Rails **26** may be joined to the bottom perimeter of the cabinet **24** and provide a foundation for the HVAC unit **12**. In certain embodiments, the rails **26** may provide access for a forklift and/or overhead rigging to facilitate installation and/or removal of the HVAC unit **12**. In some embodiments, the rails **26** may fit onto “curbs” on the roof to enable the HVAC unit **12** to provide air to the ductwork **14** from the bottom of the HVAC unit **12** while

blocking elements such as rain from leaking into the building **10**.

(29) The HVAC unit **12** includes heat exchangers **28** and **30** in fluid communication with one or more refrigeration circuits. Tubes within the heat exchangers **28** and **30** may circulate refrigerant, such as R-**410A**, through the heat exchangers **28** and **30**. The tubes may be of various types, such as multichannel tubes, conventional copper or aluminum tubing, and so forth. Together, the heat exchangers **28** and **30** may implement a thermal cycle in which the refrigerant undergoes phase changes and/or temperature changes as it flows through the heat exchangers **28** and **30** to produce heated and/or cooled air. For example, the heat exchanger **28** may function as a condenser where heat is released from the refrigerant to ambient air, and the heat exchanger **30** may function as an evaporator where the refrigerant absorbs heat to cool an air stream. In other embodiments, the HVAC unit **12** may operate in a heat pump mode where the roles of the heat exchangers **28** and **30** may be reversed. That is, the heat exchanger **28** may function as an evaporator and the heat exchanger **30** may function as a condenser. In further embodiments, the HVAC unit **12** may include a furnace for heating the air stream that is supplied to the building **10**. While the illustrated embodiment of FIG. **2** shows the HVAC unit **12** having two of the heat exchangers **28** and **30**, in other embodiments, the HVAC unit **12** may include one heat exchanger or more than two heat exchangers.

(30) The heat exchanger **30** is located within a compartment **31** that separates the heat exchanger **30** from the heat exchanger **28**. Fans **32** draw air from the environment through the heat exchanger **28**. Air may be heated and/or cooled as the air flows through the heat exchanger **28** before being released back to the environment surrounding the HVAC unit **12**. A blower assembly **34**, powered by a motor **36**, draws air through the heat exchanger **30** to heat or cool the air. The heated or cooled air may be directed to the building **10** by the ductwork **14**, which may be connected to the HVAC unit **12**. Before flowing through the heat exchanger **30**, the conditioned air flows through one or more filters **38** that may remove particulates and contaminants from the air. In certain embodiments, the filters **38** may be disposed on the air intake side of the heat exchanger **30** to prevent contaminants from contacting the heat exchanger **30**.

(31) The HVAC unit **12** also may include other equipment for implementing the thermal cycle. Compressors **42** increase the pressure and temperature of the refrigerant before the refrigerant enters the heat exchanger **28**. The compressors **42** may be any suitable type of compressors, such as scroll compressors, rotary compressors, screw compressors, or reciprocating compressors. In some embodiments, the compressors **42** may include a pair of hermetic direct drive compressors arranged in a dual stage configuration **44**. However, in other embodiments, any number of the compressors **42** may be provided to achieve various stages of heating and/or cooling. Additional equipment and devices may be included in the HVAC unit **12**, such as a solid-core filter drier, a drain pan, a disconnect switch, an economizer, pressure switches, phase monitors, and humidity sensors, among other things.

(32) The HVAC unit **12** may receive power through a terminal block **46**. For example, a high voltage power source may be connected to the terminal block **46** to power the equipment. The operation of the HVAC unit **12** may be governed or regulated by a control board **48**. The control board **48** may include control circuitry connected to a thermostat, sensors, and alarms. One or more of these components may be referred to herein separately or collectively as the control device **16**. The control circuitry may be configured to control operation of the equipment, provide alarms, and monitor safety switches. Wiring **49** may connect the control board **48** and the terminal block **46** to the equipment of the HVAC unit **12**.

(33) FIG. **3** illustrates a residential heating and cooling system **50**, also in accordance with present techniques. The residential heating and cooling system **50** may provide heated and cooled air to a residential structure, as well as provide outside air for ventilation and provide improved indoor air quality (IAQ) through devices such as ultraviolet lights and air filters. In the illustrated embodiment, the residential heating and cooling system **50** is a split HVAC system. In general, a

residence **52** conditioned by a split HVAC system may include refrigerant conduits **54** that operatively couple the indoor unit **56** to the outdoor unit **58**. The indoor unit **56** may be positioned in a utility room, an attic, a basement, and so forth. The outdoor unit **58** is typically situated adjacent to a side of residence **52** and is covered by a shroud to protect the system components and to prevent leaves and other debris or contaminants from entering the unit. The refrigerant conduits **54** transfer refrigerant between the indoor unit **56** and the outdoor unit **58**, typically transferring primarily liquid refrigerant in one direction and primarily vaporized refrigerant in an opposite direction.

(34) When the system shown in FIG. **3** is operating as an air conditioner, a heat exchanger **60** in the outdoor unit **58** serves as a condenser for re-condensing vaporized refrigerant flowing from the indoor unit **56** to the outdoor unit **58** via one of the refrigerant conduits **54**. In these applications, a heat exchanger **62** of the indoor unit functions as an evaporator. Specifically, the heat exchanger **62** receives liquid refrigerant, which may be expanded by an expansion device, and evaporates the refrigerant before returning it to the outdoor unit **58**.

(35) The outdoor unit **58** draws environmental air through the heat exchanger **60** using a fan **64** and expels the air above the outdoor unit **58**. When operating as an air conditioner, the air is heated by the heat exchanger **60** within the outdoor unit **58** and exits the unit at a temperature higher than it entered. The indoor unit **56** includes a blower or fan **66** that directs air through or across the indoor heat exchanger **62**, where the air is cooled when the system is operating in air conditioning mode. Thereafter, the air is passed through ductwork **68** that directs the air to the residence **52**. The overall system operates to maintain a desired temperature as set by a system controller. When the temperature sensed inside the residence **52** is higher than the set point on the thermostat, or the set point plus a small amount, the residential heating and cooling system **50** may become operative to refrigerate additional air for circulation through the residence **52**. When the temperature reaches the set point, or the set point minus a small amount, the residential heating and cooling system **50** may stop the refrigeration cycle temporarily.

(36) The residential heating and cooling system **50** may also operate as a heat pump. When operating as a heat pump, the roles of heat exchangers **60** and **62** are reversed. That is, the heat exchanger **60** of the outdoor unit **58** will serve as an evaporator to evaporate refrigerant and thereby cool air entering the outdoor unit **58** as the air passes over the outdoor heat exchanger **60**. The indoor heat exchanger **62** will receive a stream of air blown over it and will heat the air by condensing the refrigerant.

(37) In some embodiments, the indoor unit **56** may include a furnace system **70**. For example, the indoor unit **56** may include the furnace system **70** when the residential heating and cooling system **50** is not configured to operate as a heat pump. The furnace system **70** may include a burner assembly and heat exchanger, among other components, inside the indoor unit **56**. Fuel is provided to the burner assembly of the furnace system **70** where it is mixed with air and combusted to form combustion products. The combustion products may pass through tubes or piping in a heat exchanger, separate from heat exchanger **62**, such that air directed by the blower or fan **66** passes over the tubes or pipes and extracts heat from the combustion products. The heated air may then be routed from the furnace system **70** to the ductwork **68** for heating the residence **52**.

(38) FIG. **4** is an embodiment of a vapor compression system **72** that can be used in any of the systems described above. The vapor compression system **72** may circulate a refrigerant through a circuit starting with a compressor **74**. The circuit may also include a condenser **76**, an expansion valve(s) or device(s) **78**, and an evaporator **80**. The vapor compression system **72** may further include a control panel **82** that has an analog to digital (A/D) converter **84**, a microprocessor **86**, a non-volatile memory **88**, and/or an interface board **90**. The control panel **82** and its components may function to regulate operation of the vapor compression system **72** based on feedback from an operator, from sensors of the vapor compression system **72** that detect operating conditions, and so forth.

(39) In some embodiments, the vapor compression system **72** may use one or more of a variable speed drive (VSDs) **92**, a motor **94**, the compressor **74**, the condenser **76**, the expansion valve or device **78**, and/or the evaporator **80**. The motor **94** may drive the compressor **74** and may be powered by the variable speed drive (VSD) **92**. The VSD **92** receives alternating current (AC) power having a particular fixed line voltage and fixed line frequency from an AC power source, and provides power having a variable voltage and frequency to the motor **94**. In other embodiments, the motor **94** may be powered directly from an AC or direct current (DC) power source. The motor **94** may include any type of electric motor that can be powered by a VSD or directly from an AC or DC power source, such as a switched reluctance motor, an induction motor, an electronically commutated permanent magnet motor, or another suitable motor.

(40) The compressor **74** compresses a refrigerant vapor and delivers the vapor to the condenser **76** through a discharge passage. In some embodiments, the compressor **74** may be a centrifugal compressor. The refrigerant vapor delivered by the compressor **74** to the condenser **76** may transfer heat to a fluid passing across the condenser **76**, such as ambient or environmental air **96**. The refrigerant vapor may condense to a refrigerant liquid in the condenser **76** as a result of thermal heat transfer with the environmental air **96**. The liquid refrigerant from the condenser **76** may flow through the expansion device **78** to the evaporator **80**.

(41) The liquid refrigerant delivered to the evaporator **80** may absorb heat from another air stream, such as a supply air stream **98** provided to the building **10** or the residence **52**. For example, the supply air stream **98** may include ambient or environmental air, return air from a building, or a combination of the two. The liquid refrigerant in the evaporator **80** may undergo a phase change from the liquid refrigerant to a refrigerant vapor. In this manner, the evaporator **80** may reduce the temperature of the supply air stream **98** via thermal heat transfer with the refrigerant. Thereafter, the vapor refrigerant exits the evaporator **80** and returns to the compressor **74** by a suction line to complete the cycle.

(42) In some embodiments, the vapor compression system **72** may further include a reheat coil in addition to the evaporator **80**. For example, the reheat coil may be positioned downstream of the evaporator relative to the supply air stream **98** and may reheat the supply air stream **98** when the supply air stream **98** is overcooled to remove humidity from the supply air stream **98** before the supply air stream **98** is directed to the building **10** or the residence **52**.

(43) It should be appreciated that any of the features described herein may be incorporated with the HVAC unit **12**, the residential heating and cooling system **50**, or other HVAC systems.

Additionally, while the features disclosed herein are described in the context of embodiments that directly heat and cool a supply air stream provided to a building or other load, embodiments of the present disclosure may be applicable to other HVAC systems as well. For example, the features described herein may be applied to mechanical cooling systems, free cooling systems, chiller systems, or other heat pump or refrigeration applications.

(44) As set forth above, embodiments of the present disclosure are directed to a collapsible air hood assembly that is configured to couple to an HVAC unit and transition from a collapsed configuration to a deployed configuration and vice versa. The air hood assembly may be configured to couple to the HVAC unit **12** of FIG. **1**, the outdoor unit **58** illustrated in FIG. **3**, and/or any other suitable HVAC unit. Additionally, the air hood assembly may transition from the deployed configuration to the collapsed configuration when transportation of the HVAC unit is desired. Therefore, the air hood assembly may remain coupled to the HVAC unit during transportation, thereby reducing or eliminating separate shipping costs for each of the air hood assembly and the HVAC unit. Additionally, adjusting the air hood assembly between the collapsed configuration and the deployed configuration may be less time-consuming when compared to a traditional installation of the air hood assembly. As such, an assembly time of the HVAC unit may also be reduced.

(45) FIG. **5** is a perspective view of a side **102** (e.g., lateral side or surface) of an HVAC unit **100**,

illustrating an embodiment of an air hood assembly **200** (e.g., hood assembly, intake hood assembly) having an inlet hood **202** in a deployed configuration. As shown in the illustrated embodiment of FIG. 5, the HVAC unit **100** may include a housing **104** configured to support and secure the air hood assembly **200** on the side **102** of the HVAC unit **100**. The housing **104** may also support and/or couple to other various components of the HVAC unit **100**. For example, the housing **104** may include a first sub-frame **106** (e.g., frame, frame member, support structure) to which the inlet hood **202** may be secured, and a second sub-frame **108** (e.g., frame, frame member, support structure) to which an exhaust hood **204** of the HVAC unit **100** may be secured. The exhaust hood **204** may be configured to protect or cover a louver **205**, which may be configured to regulate a flow of exhaust air discharged from the HVAC unit **100**. It should be noted that a similar louver may be present within the inlet hood **202** and may be configured to regulate a flow of intake air received by the HVAC unit **100**.

(46) The air hood assembly **200** may include a top panel **206** (e.g., top panel assembly, top cover) and a filter assembly **208**, each rotatably coupled to the first sub-frame **106**. The top panel **206** includes a first panel portion **210** and a second panel portion **212** that may be adjustably coupled to one another. For example, during a transition of the air hood assembly **200** from a deployed configuration to a collapsed configuration, the second panel portion **212** may slide or translate relative to the first panel portion **210** to enable the first panel portion **210** to substantially cover and contain the filter assembly **208** within and/or against the side **102** of the housing **104** of the HVAC unit **100**, as described in greater detail below. The filter assembly **208** may include a frame **213** (e.g., support structure) configured to support filters of the air hood assembly **200**. For example, the frame **213** may include a first filter frame **214** (e.g., sub-frame) rotatably coupled to the top panel **206** and a second filter frame **216** (e.g., sub-frame) rotatably coupled to the first sub-frame **106** of the housing **104**. The first filter frame **214** and the second filter frame **216** may each be configured to support one or more filters **215**, **217**. The filters **215**, **217** may be configured to block particulates and/or contaminants from entering an interior of the HVAC unit **100** from the environment surrounding the HVAC unit **100**, such as during operation of the HVAC unit **100**. The filter assembly **208** may also include a mesh **218** to configured to secure and hold the one or more filters **215**, **217** in the first filter frame **214** and the second filter frame **216**.

(47) The first filter frame **214** and the second filter frame **216** may also be rotatably coupled to one another by one or more hinge joints **219** (e.g., hinges, pivot connections, etc.). The one or more hinge joints **219** may be configured to couple the first and second filter frames **214**, **216** together to facilitate the transition of the air hood assembly **200** from a deployed configuration to a collapsed configuration, as described in greater detail with reference to FIGS. 8 and 11A. As discussed in detail herein, the filter assembly **208** may be positioned over an opening (e.g., air inlet) of the HVAC unit **100**. Accordingly, air may be drawn through the filters **215**, **217** and into an interior of the HVAC unit **100** via a fan or blower of the HVAC unit **100**. The air may be directed through the HVAC unit **100** and may pass over one or more heat exchangers within the HVAC unit **100** to condition the air.

(48) In the deployed configuration, the first filter frame **214** and the second filter frame **216** may be extended and/or expanded out towards the environment and away from the HVAC unit **100** such that the first filter frame **214** and the second filter frame **216** are aligned with one another (e.g., side-by-side, adjacent one another) in an extended position. In the deployed configuration, a gap **222** may be formed between the first filter frame **214** and the second filter frame **216**. The gap **222** may be occluded by a flexible material (e.g., a filter and/or cloth strip) configured to span the gap **222** and between the first filter frame **214** and the second filter frame **216**. The gap **222** may be formed in the deployed configuration as a result of the spatial relationship of the first filter frame **214** relative to the second filter frame **216** (e.g., via the hinge joints **219**), and the spatial relationship may facilitate transition of the air hood assembly **200** from the deployed configuration to the collapsed configuration and vice versa, as described in greater detail below. In the deployed

configuration, the air hood assembly **200** may further incorporate a pair of side panels **224** (e.g., on opposite sides of the inlet hood **202**) configured to couple to, secure, and/or support the top panel **206** and the filter assembly **208** in the extended position while in the air hood assembly **200** is in the deployed configuration. As shown in the illustrated embodiment of FIG. 5, the side panels **224** may extend laterally outward from the side **102** of the HVAC unit **100**. For example, the side panels **224** may each include triangular geometry or profile having a first edge **225** configured to couple to and/or abut the housing **104** of the HVAC unit **100**. Additionally, a second edge **226** of each side panel **224** may be configured to couple to the top panel **206**. Further still, the side panels **224** may each include a third edge **227** configured to couple to and/or abut the filter assembly **208**. The third edges **227** of the side panels **224** may also be configured to secure or lock the filter assembly **208** in the extended position while the air hood assembly **200** is in the deployed configuration. As discussed further below, the side panels **224** may be removed from the air hood assembly **200** to facilitate transition of the air hood assembly **200** from the deployed configuration to the collapsed configuration.

(49) FIG. 6 is a perspective view of an embodiment of the air hood assembly **200** in the deployed configuration. The inlet hood **202** is configured to block water and/or other contaminants from entering the interior of the HVAC unit **100** and/or otherwise through the filters **215**, **217**. As discussed above, the top panel **206** may include the first panel portion **210** and the second panel portion **212** and may be disposed at an oblique angle **150** relative to the side **102** of the HVAC unit **100** such that the top panel **206** may provide protection for components of the air hood assembly **200** positioned below the top panel **206** (e.g., the filter assembly **208**) and/or within the housing of the HVAC system **100** (e.g., a louver). The first panel portion **210** may have a top edge **231** (e.g., a base edge, first edge), a bottom edge **232** (e.g., a distal edge, second edge) opposite the top edge **231** and having a lip **300**, and a pair of lateral edges **233**, **234**. The second panel portion **212** may also have a top edge **235** (e.g., overlapping edge, first edge), a bottom edge **236** (e.g., a distal edge, second edge) opposite the top edge **235** and having a lip **302**, and a pair of lateral edges **237**, **238**. As illustrated, the top edge **235** of the second panel portion **212** may be positioned underneath or behind the first panel portion **210** to enable the second panel portion **212** to slide relative to the first panel portion **210**. In the collapsed configuration of the air hood assembly **200**, the second panel portion **212** may be retracted behind and substantially covered (e.g., via overlap) by the first panel portion **210**. The first panel portion **210** may be configured to maintain its shape or geometry while transitioning from a deployed configuration to a collapsed configuration. For example, regardless of configuration (e.g., deployed, transitioning, collapsed) of the air hood assembly **200**, the first panel portion **210** may remain fixed relative to the second panel portion **212**, and the second panel portion **212** may slide or translate towards or away from the first panel portion **210**. That is, the lip **302** of the second panel portion **212** may move towards the lip **300** of the first panel portion **210** when the air hood assembly **200** transitions from the deployed configuration to the collapsed configuration. For example, the second panel portion **212** may translate relative to the first panel portion **210** until the bottom edge **236** and the lip **302** of the second panel portion **212** contact and/or are adjacent to the bottom edge **232** and the lip **300** of the first panel portion **210**. In the collapsed configuration, the first panel portion **212** may substantially cover and/or overlap with the second panel portion **210**, as illustrated in FIG. 12. When the air hood assembly **200** transitions from the collapsed configuration to the deployed configuration, the bottom edge **236** and the lip **302** of the second panel portion **212** may extend away from the bottom edge **232** and the lip **300** of the first panel portion **210**. In this manner, the top panel **206** enables the filter assembly **208** to extend out into the deployed configuration (e.g., with filter frames **214**, **216** side-by-side). It should be noted that, as used herein, the term “top” is not limited to orientations or spatial relationships whereby an element (e.g., the top panel **206**) is located above (e.g., vertically above) other elements (e.g., relative to gravity). That is, in some configurations (e.g., the collapsed configuration), the top panel **206** may be the most outward element relative to the housing **104** such that the top panel **206**

substantially covers and contains other components of the air hood assembly **200** within the housing **104**, as described in greater detail below.

(50) As illustrated, the first edges **225** of the side panels **224** may be coupled to the housing **104** via one or more fasteners **304**. The fasteners **304** may be pins or screws that are readily removed to facilitate the transition of the air hood assembly **200** from the deployed configuration to the collapsed configuration. The second edge **226** of the side panel **224** may be coupled to the top panel **206** via one or more fasteners **305**. The fasteners **305** may be similar to fasteners **304** to enable ready removal of the side panels **224** to enable transition of the air hood assembly **200** to the collapsed configuration. The third edge **227** of the side panel **224** may be positioned above the filter assembly **208** (e.g., relative to gravity) in the deployed configuration in order to bias (e.g., via a force **310** in an outward and/or downward direction) and/or retain (e.g., secure) the filter assembly **208** in the extended position.

(51) As described above, the filter assembly **208** may include the first filter frame **214**, the second filter frame **216**, filters **215**, **217**, and the mesh **218**. The first filter frame **214** may have a top side **240** (e.g., distal side, first side), a bottom side **241** (adjoining side, second side), and a pair of lateral sides **242**, **243**. The second filter frame **216** may also have a top side **244** (e.g., adjoining side, first side), a bottom side **245** (e.g., mounting side, second side), and a pair of lateral sides **246**, **247**. When the filter assembly **208** is in the extended position, the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** may be separated by the gap **222**. The bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** are also rotatably coupled to one another such that the first filter frame **214** and the second filter frame **216** may angularly rotate towards and/or away from one another during transition of the air hood assembly **200** between the deployed configuration and the collapsed configuration via the hinge joints **219**. The spatial relationship of the first filter frame **214** and the second filter frame **216** that results in the gap **222** in the deployed configuration may also provide additional space for the first filter frame **214** and the second filter frame **216** to rotate relative to one another. The gap **222** may be enclosed by a flexible material, such as a filter cloth strip, configured to fit within (e.g., fill, span) the gap **222**, thereby blocking air flow between the first filter frame **214** and the second filter frame **216** and into the inlet hood **202**.

(52) The top side **240** of the first filter frame **214** may be rotatably coupled to the bottom edge **236** of the second panel portion **212**. Further, each of the sides **240**, **241**, **242**, **243**, **244**, **245**, **246**, and **247** of the first filter frame **214** and the second filter frame **216** may include or be formed from C-channels configured to facilitate installation of the filters **215**, **217** therein. It should be noted that each of the filters **215**, **217** may each include multiple individual filters. When the filter **215** is installed within the first filter frame **214**, a gap (e.g., space, void) may extend between an edge of the filter **215** and the top side **240** of the first filter frame **214**. This gap (e.g., within the C-channel forming the top side **240**) may provide room for the filter **215** to be adjusted or manipulated during replacement of one or more of the filters **215** (e.g., without disassembling the filter assembly **208**), as described in greater detail with reference to FIG. **13**. It should be noted that each of the filter frame sides **240**, **241**, **242**, **243**, **244**, **245**, **246**, and **247**, which may be formed or defined by one or more C-channels, may provide a gap similar to that described above between the side of the corresponding filter frame **214**, **216** and a respective side of the corresponding filter **215**, **217** to facilitate installation and/or removal of the filters **215**, **217**.

(53) As discussed above, various components of the air hood assembly **200** may be coupled to the first sub-frame **106**. The first sub-frame **106** may have a top portion **111** (e.g., first portion), a bottom portion **112** (e.g., second portion) opposite the top portion **111** relative to an air inlet protected by the air hood assembly **200**, a first side portion **113**, and a second side portion **114** opposite the first side portion **113** relative to the air inlet protected by the air hood assembly **200**. The top portion **111** may also include a top panel retainer **115** (e.g., clasp, clamp, bracket), which is described further below. Each portion **111**, **112**, **113**, and **114** of the first sub-frame **106** may be

configured to couple to and secure components of the air hood assembly **200** to the housing **104** of the HVAC unit **100**. For example, the bottom side **245** of the second filter frame **216** may be rotatably coupled to the bottom portion **112** of the first sub-frame **106** via hinge joints **220**, and the top edge **231** of the first panel portion **210** may be rotatably coupled to the first portion **111** of the first sub-frame **106** via the top panel retainer **115**. Further, while in the deployed configuration, the first edges **225** of the side panels **224** may be coupled to the first side portion **113** and the second side portion **114**, respectively, via the fasteners **304**.

(54) FIG. 7A is a side perspective view of an embodiment of the top panel retainer **115**, which may be configured to facilitate the securement of the top edge **231** of the first panel portion **210** to the housing **104** of the HVAC system **100** in the deployed configuration. For example, the top panel retainer **115** may be secured to the top portion **111** of the first sub frame **106** via one or more fasteners **306**. The one or more fasteners **306** may extend through the top portion **111** of the first sub frame **106** and into the top panel retainer **115**, thereby securing the top panel retainer **115** to the housing **104**. The top edge **231** of the first panel portion **210** may have a lip **310** that extends in a generally vertical direction when the air hood assembly **200** is in the deployed configuration, and the top panel retainer **115** may also have a lip **116** that extends in a generally horizontal direction.

(55) As illustrated and discussed above, fasteners **304** may extend through the first edge **225** of the side panel **224** and into the second side portion **114** of the first sub-frame **106** to couple and secure the side panel **224** to the housing **104**. Further, fasteners **305** may extend through the first panel portion **210** and into the second edge **226** of the side panel **224**, thereby coupling and securing the first panel portion **210** to the side panel **224**. When the fasteners **304**, **305** are removed along with the side panels **224** (e.g., to transition the air hood assembly **200** from the deployed configuration to the collapsed configuration), the top panel **206** may rotate or pivot towards the housing **104** in a generally downward and/or inward direction **400** relative to the housing **104**, and the lip **310** may rotate or pivot towards the lip **116** of the top panel retainer **115** in a generally downward and/or inward direction **350** relative to the lip **116**. In the collapsed configuration, the top panel **206** extends along and generally flush with the housing **104**, as discussed above. Additionally, in this position of the top panel **206**, the lip **116** may extend along and rest against (e.g., above and/or on top of, relative to gravity) the lip **310**, thereby securing the top panel **206** within the top panel retainer **115** in the collapsed configuration, as discussed in greater detail below with reference to FIG. 7B.

(56) FIG. 7B is an exploded view of an embodiment of the top panel retainer **115** with the top panel **206** of the air hood assembly **200** in the collapsed configuration. When in the collapsed configuration, the lip **310** of the first panel portion **210** may be positioned above the lip **116** of the top panel retainer **115** relative to gravity. As illustrated, the lip **310** of the first panel portion **210** may rest against the lip **116** of the top panel retainer **115** in the collapsed configuration, such that the lip **116** of the top panel retainer **115** supports the top panel **206** via engagement with the lip **310** of the first panel portion **210**. In this way, the top panel retainer **115** restricts or blocks the top panel **206** from moving downward relative to the top panel retainer **115**. For example, the lip **116** of the top panel retainer **115** may be configured to support a weight of the top panel **206** while in the collapsed configuration. Further, the lip **116** may be configured to restrict the top panel **206** from rotating further inward, relative to the housing **104**, upon reaching the collapsed configuration. Furthermore, as shown, the lip **310** and the lip **116** are dimensioned such that the lips **310**, **116** substantially overlap with and contact one another in the collapsed configuration of the air hood assembly **200**. For example, a full length or a substantially full length of the lip **310** rests on the lip **116**, and the lip **116** extends toward the housing **104** beneath (e.g., relative to gravity) the full length or substantially full length of the lip **310**. Thus, the top panel retainer **115** may block or mitigate inadvertent dislodging of the top panel **206** from the top panel retainer **115**.

(57) Turning now to FIG. 8A, a schematic view illustrating transition of an embodiment of the air hood assembly **200** between a deployed configuration and a collapsed configuration, is shown.

During transition, the side panels **224** may be removed to enable rotation and/or pivoting of the top panel **206** relative to the housing **104**, as enabled by the lip **116** of the top panel retainer **115** and the lip **310** of the first panel portion **210**. As discussed above, the top panel **206** may include the first panel portion **210** and the second panel portion **212**. During transition from the deployed configuration to the collapsed configuration, the top panel **206** may rotate in the downward and/or inward direction **400** relative to the housing **104** such that the filter assembly **208** may collapse and be protected and contained within the housing **104**. Further, during transition, the second panel portion **212** may move in a generally inward direction **402** relative to the first panel portion **210**. Because the second panel portion **212** is coupled to the first filter frame **214**, when the second panel portion **212** slides inward relative to the first panel portion **210**, the first filter frame **214** also moves inward towards the housing **104** of the HVAC unit **100** such that when the air hood assembly **200** is fully collapsed, the filter assembly **208** is covered and contained within the housing **104** by the top panel **206**. During transition from the collapsed configuration to the deployed configuration, the top panel **206** may rotate in an outward and/or upward direction **500** relative to the housing **104**. Further, the second panel portion **212** may move in a generally outward direction **502** relative to the first panel portion **210**. In turn, the first filter frame **214** may move away from the housing **104** to provide additional space for the filter assembly **208** to move (e.g., rotate and/or pivot) into the extended position. To facilitate transition, the first filter frame **214** and the second filter frame **216** may be coupled together via the one or more hinge joints **219**. The one or more hinge joints **219** may be configured to enable rotational movement between the first filter frame **214** and the second filter frame **216**. Further, one or more additional hinge joints **220** may couple the second filter frame **216** to the first sub-frame **106**. The one or more additional hinge joints **220** may enable rotational movement between the bottom side **245** of the second filter frame **216** and the bottom portion **112** of the first sub-frame **106**. Such features are described in greater detail with reference to FIG. **11A** and FIG. **11B**.

(58) As discussed above, the first filter frame **214** and the second filter frame **216** are rotatably coupled to one another by the one or more hinge joints **219**. During transition from the deployed configuration to the collapsed configuration, the first filter frame **214** and the second filter frame **216** may rotate towards each other (e.g., about the one or more hinge joints **219**) until the air hood assembly **200** is in the collapsed configuration. When the air hood assembly **200** is fully collapsed, the first filter frame **214** and the second filter frame **216** overlap with one another (e.g., in a side-by-side or adjacent arrangement) and may be substantially contained against the housing **104** (e.g., covered) by the top panel **206**. For example, the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** may be rotatably coupled to one another via the hinge joints **219** such that the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** may move in a generally inward and/or upward direction **404** relative to the housing **104** when the air hood assembly **200** transitions from the deployed configuration to the collapsed configuration. As the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** pivot or rotate and move in the inward and/or upward direction **404**, an inward surface **270** of the first filter frame **214** may pivot an inward and/or upward direction **274** relative to the top panel **206**. Further, an inward surface **280** of the second filter frame **216** may pivot in an inward and/or upward direction **284** relative to the first sub-frame **106** of the housing **104**. Thus, during transition from the deployed configuration to the collapsed configuration, the first filter frame **214** and the second filter frame **216** may rotate or pivot towards one another until they are substantially flush against one another and arranged to overlap one another. For example, in the collapsed configuration and relative to the housing **104**, the first panel portion **210** is external to the second panel portion **212**, the second panel portion **212** is external to the first filter frame **214**, and the first filter frame **214** is external to the second filter frame **216**. Thus, the first panel portion **210** substantially covers and contains the second panel portion **210** and the first and second filter frames **214**, **216** against the housing **104** in the collapsed configuration. It should be noted

that, as the second panel portion **212** moves in the inward direction **402** relative to the first panel portion **210**, the top side **240** of the first filter frame **214** may move in a similar direction due to the coupling between the second panel portion **212** and the first filter frame **214**.

(59) During transition from the collapsed configuration to the deployed configuration, the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** may move in a generally outward and/or downward direction **504** relative to the housing **104** to enable transition of the filter assembly **208** into the extended position associated with the deployed configuration of the air hood assembly **200**. As the bottom side **241** of the first filter frame **214** and the top side **244** of the second filter frame **216** move in the outward and/or downward direction **504** relative to the housing **104**, an outward surface **272** of the first filter frame **214** may rotate in an outward and/or downward direction **276** relative to the top panel **206** and an outward surface **282** of the second filter frame **216** may rotate in an outward and/or downward direction **286** relative to the first sub frame **106** of the housing **104**. Thus, during transition from the collapsed configuration to the deployed configuration, the first filter frame **214** and the second filter frame **216** may rotate away from one another such that, when extended, the first filter frame **214** and the second filter frame **216** are aligned with one another (e.g., end-to-end) in the extended position. It should be noted that, as the second panel portion **212** moves outward relative to the first panel portion **210**, the top side **240** of the first filter frame **214** may move in a similar direction due to the coupling between the second panel portion **212** and the first filter frame **214**.

(60) FIG. **8B** a perspective view of an embodiment of the air hood assembly **200** transitioning from a deployed configuration to a collapsed configuration. As described above, the filter assembly **208** may be configured to fold or collapse towards the housing **104** when transitioning from the deployed configuration to the collapsed configuration. In the illustrated embodiment, as the filter assembly **208** collapses towards the first sub-frame **106** of the housing **104**, the first filter frame **214** may be configured to sit generally flush against the top panel **206** such that the top panel **206** may cover (e.g., be disposed external to, relative to the housing **104**) a portion of the first filter frame **214** and the filter assembly **208**. Further, because the bottom edge **236** of the second panel portion **212** is rotatably coupled to the top side **240** of the first filter frame **214**, as the second panel portion **212** slides or translates relative to the first panel portion **210**, the first filter frame **214** may also slide or translate relative to the first panel portion **210** towards the housing **104**. Thus, when in the collapsed configuration, the first panel portion **210** substantially covers and contains the filter assembly **208** within and/or against the housing **104**.

(61) FIG. **9** is a top-perspective view of an embodiment of the air hood assembly **200** in a deployed configuration. In the deployed configuration, the side panels **224** may be coupled to the first sub-frame **106** via fasteners **304** and to the first panel portion **210** via fasteners **305**. Once the fasteners **304**, **305** and side panels **224** are removed, the top panel **206** may rotate towards the first sub-frame **106** to the collapsed configuration. As discussed above, the top edge **235** of the second panel portion **212** may be positioned underneath the bottom edge **232** of the first panel portion **210**, and the second panel portion **212** may be configured to slide or translate relative to the first panel portion **210** to enable the air hood assembly **200** to transition to the collapsed configuration. To facilitate this movement, the second panel portion **212** may have a slot **320** formed in each of the lateral edges **237**, **238** of the second panel portion **212**. The slot **320** may be configured to guide or direct the second panel portion **212** towards the first panel portion **210** via one or more bolts, as described in greater detail below.

(62) For example, FIG. **10A** shows an exploded view of an embodiment of the slot **320** configured to facilitate movement of the second panel portion **212** relative to the first panel portion **210**. The slot **320** may span across a length **322** of the lateral edge **237** of the second panel portion **212**. As discussed above, the top edge **235** of the second panel portion **212** may be positioned underneath the bottom edge **232** of the first panel portion **210**, thereby enabling the second panel portion **212** to translate in either the inward direction **402** or outward direction **502** relative to the first panel

portion **210**, such as along a first axis **500**. To facilitate the sliding of the second panel portion **212** relative to the first panel portion **210**, a first bolt **330** (e.g., pin, fastener, etc.) may extend through the first panel portion **210** at the bottom edge **232** of the first panel portion **210** and into the slot **320**. The first bolt **330** may be configured to couple the first panel portion **210** to the second panel portion **212** while also enabling movement between the first panel portion **210** and the second panel portion **212**. That is, the first bolt **330** may be tight enough to secure the first panel portion **210** to the second panel portion **212**, but loose enough to enable movement between the two portions **210**, **212**. A second bolt **332** (e.g., pin, fastener, etc.) may be positioned within the slot **320** along or at the bottom edge **236** of the second panel portion **212**. One or more fasteners **308** may be configured to secure the second panel portion **212** to the side panel **224** while the air hood assembly **200** is in the deployed configuration. When transitioning from the deployed configuration to the collapsed configuration, the fasteners **305**, **308** and the side panels **224** may be removed, and the first bolt **330** and the second bolt **332** may move along the slot **320** towards the first bolt **330** within the slot **320** to guide the second panel portion **212** in the upward direction **402** along the first axis **500**. For example, because the first panel portion **210** remains fixed relative to the second panel portion **212**, the first bolt **330** may also remain in a fixed position along the bottom edge **232** of the first panel portion **210**. As the second panel portion **212** slides towards the first panel portion **210** along the first axis **500**, the first bolt **330** may be contained within the slot **320** thereby limiting movement along a second axis **510**. It should be noted that a similar (e.g., additional) slot **320** may be formed along the lateral edge **238** of the second panel portion **212** and may be configured to function similar to the slot **320** described above.

(63) FIG. **10B** is an exploded view of an embodiment of the coupling between the first panel portion **210** and the second panel portion **212** via the first bolt **330**. As discussed above, the fastener **305** may be configured to couple the first panel portion **210** to the side panel **224** and may be readily removed to enable the second panel portion **210** to slide relative to the first panel portion **210** when the air hood assembly **200** transitions to the collapsed configuration. As illustrated, the bottom edge **232** of the first panel portion **210** is positioned above and/or external to the top edge **235** of the second panel portion **212**, and the first bolt **330** may be configured to extend through the bottom edge **232** of the first panel portion **210** and into the second panel portion **212** via the slot **320**. The first bolt **330** may be retained within the slot **320** by a nut **331** that is configured to secure the coupling between the first panel portion **210** and the second panel portion **212**, as described above. That is, the nut **331** may be tight enough to couple the first panel portion **210** to the second panel portion **212** to one another, but loose enough to enable movement of the second panel portion **212** relative to the first panel portion **210** along the axis **500**. For example, the first bolt **330** may extend through and be contained within the slot **320** such that as the second panel portion **212** may slide relative to the first panel portion **210**, and movement (e.g., lateral movement) that is not along the axis **500** may be limited by the first bolt **330** retained within the slot **320**. It should be noted that the lateral edge **238** of the second panel portion **212** may have a similar slot and fastener configuration as that discussed above.

(64) As discussed above, one or more fasteners may be used to rotatably couple various components of the air hood assembly **200** to one another. FIG. **11A** is a perspective view of an embodiment of the hinge joint **219** configured to rotatably couple the first filter frame **214** and the second filter frame **216**. As discussed above, the gap **222** may be formed between the first filter frame **214** and the second filter frame **216** to provide additional space for the filter frames **214**, **216** to rotate relative to one another. The hinge joint **219** may include a joining plate **600** (e.g., a hinge plate) including a first hole **601** and a second hole **602**. The first filter frame **214** may be rotatably coupled to the joining plate **600** via a first rivet **603** (e.g., pin) extending through the first hole **601** and through the lateral side **242** of the first filter frame **214**. The second filter frame **216** may be rotatably coupled to the joining plate **600** via a second rivet **604** (e.g., pin) extending through the second hole **602** and through the lateral side **246** of the second filter frame **216**. By coupling the

first filter frame **214** and the second filter frame **216** to the joining plate **600**, the filter frames **214**, **216** may rotate relative to each other when the air hood assembly **200** transitions between the deployed configuration and the collapsed configuration. It should be noted that the lateral side **243** of the first filter frame **214** and the lateral side **247** of the second filter frame **216** (illustrated in FIG. **6**) may be coupled to one another by a similar hinge joint **219**. Further, in some embodiments, the rivets **603**, **604** may be self-tapping fastening screws, pins, or other suitable element about which the first filter frame **214** and/or second filter frame **216** may rotate.

(65) FIG. **11B** is a perspective view of an embodiment of the hinge joint **220** configured to rotatably couple the second filter frame **216** to the first sub-frame **106**. The hinge joint **220** may be an L-shaped bracket (e.g., mounting flange) having a first attaching member **610** (e.g., first flange) and a second attaching member **612** (e.g., second flange). The first attaching member **610** may be configured to secure the hinge joint **220** to the first sub-frame **106** of the housing **104** via one or more bolts **620** or other mechanical fasteners. The one or more bolts **620** may secure the first attaching member **610** to the first sub-frame **106** such that the hinge joint **220** is fixedly attached to the first sub-frame **106**. The second attaching member **612** may have a hole **614** configured to facilitate the adjustable coupling between the second filter frame **216** and the first sub-frame **106**. That is, the second filter frame **214** may be rotatably coupled to the second attaching member **612** via a rivet **615** extending through the hole **614** and through the lateral side **246** of the second filter frame **216**. By coupling the second filter frame **216** to the second attaching member **612**, the second filter frame **214** is configured to rotate relative to the first sub-frame **106** of the housing **104**, thereby facilitating transition of the air hood assembly **200** between the deployed configuration and the collapsed configuration. It should be noted that a similar hinge joint **220** may be included on the lateral side **247** (illustrated in FIG. **6**) of the second filter frame **216**, and in some embodiments, the rivet **615** may include a self-tapping fastening screw, pin, or other suitable element about which the second filter frame **216** may rotate.

(66) Referring now to FIG. **12**, a perspective view of an embodiment of the air hood assembly **200** in a collapsed configuration is illustrated. As shown, the side panels **224** are removed such that the top panel **206** may rotate to rest along the housing **104** (e.g., in a generally vertical orientation) and to substantially cover and contain the filter assembly **208** against and/or within the housing **104** of the HVAC unit **100**. As discussed above, during transition from the deployed configuration to the collapsed configuration, the second panel portion **212** may slide or translate relative to the first panel portion **210**, such that the second panel portion **212** may be substantially covered by the first panel portion **210**. Indeed, the first panel portion **210** may sit generally flush against the housing **104**, thereby protecting and containing the filter assembly **208** within the housing **104**. That is, the first panel portion **210** may overlap and be disposed external to the second panel portion **212** and the filter assembly **208** contained within the housing **104**. For example, as illustrated, the bottom edge **232** of the first panel portion **210** is substantially aligned with the bottom edge **236** of the second panel portion **212**, such that the second panel portion **212** is substantially covered and/or disposed behind by the first panel portion **210** (e.g., relative to an external facing surface of the first panel portion **210**). That is, in the collapsed configuration, the first panel portion **210** may be an outermost layer of the air hood assembly **200** relative to the housing **104**, and the second panel portion **212** may be positioned internal to the first panel portion **210** relative to the housing **104**. Such a configuration enables the top panel **206** to cover and protect the filter assembly **208** within the housing **104** and further aids in installation of the air hood assembly **200**. More specifically, the filters **215**, **217** and the air hood assembly **200** may be installed with the HVAC unit **100** prior to transportation of the HVAC unit **100** (e.g., to an installation site), and the HVAC unit **100** may then be transported (e.g., shipped) without removing the filters **215**, **217** and the air hood assembly **200** because the filter assembly **208** is substantially covered and protected by the top panel **206**. Additionally, in the collapsed configuration, the air hood assembly **200** does not significantly and/or materially increase a footprint of the HVAC unit **100**, thereby reducing delivery,

transportation, and installation costs associated with the HVAC unit **100**.

(67) FIG. **13** is a schematic view of an embodiment of the filter assembly **208** with the first filter frame **214** and the second filter frame **216** arranged in the extended position (e.g., end-to-end, adjacent one another) and with the filters **215**, **217** installed. As discussed above, the first filter frame **214** and the second filter frame **216** may be separated by the gap **222** and may be coupled together via the hinge joints **219**. The filters **215**, **217** may be retained within the first and second filter frames **214**, **216** by the mesh **218**. The gap **222** may be formed in the deployed configuration as a result of the spatial relationship of the first filter frame **214** relative to the second filter frame **216** (e.g., via the hinge joints **219**), and the spatial relationship may facilitate transition of the air hood assembly **200** from the deployed configuration to the collapsed configuration and vice versa. Further, each of the sides **240**, **241**, **242**, **243**, **244**, **234**, **246**, and **247** of the filter frames **214**, **216** may include or be formed from C-channels configured to facilitate installation of the filters **215**, **217** therein. For example, when the filters **215**, **217** are installed, a first gap **700** may be formed between a top edge **800** of the filter **215** and the top side **240** of the first filter frame **214**, and a second gap **702** may be formed between a bottom edge **801** of the filter **215** and the bottom side **241** of the first filter frame **214**. Further, a third gap **704** may be formed between a first lateral edge **802** of the filter **215** and the lateral side **242** of the first filter frame **214**, and a fourth gap **706** may be formed between a second lateral edge **803** of the filter **215** and the lateral side **243** of the first filter frame **214**. Similarly, a first gap **710** may be formed between a top edge **804** of the filter **217** and the top side **244** of the second filter frame, and a second gap **712** may be formed between a bottom edge **805** of the filter **217** and the bottom side **245** of the second filter frame **216**. Further, a third gap **714** may be formed between a first lateral edge **806** of the filter **217** and the lateral side **246** of the second filter frame **216**, and a fourth gap **716** may be formed between a second lateral edge **807** of the filter **217** and the lateral side **247** of the second filter frame **216**. Each of the gaps **700**, **702**, **704**, **706**, **710**, **712**, **714**, **716** may be configured to enable the filters **215**, **217** to move relative to the first and second filter frames **214**, **216** and may provide adequate clearance for the filters **215**, **217** to be readily removed or installed as need. For example, to remove the filter **215**, the top edge **800** of the filter **215** may be transitioned into the gap **700** towards the top side **240** of the first filter frame **214** such that the bottom edge **801** of the filter **215** may no longer constrained or held by the bottom side **241** of the first filter frame **214**, thereby enabling the removal of the filter **215** from the first filter frame **214**. Similarly, to remove the filter **217**, the top edge **804** of the filter **217** may be transitioned into the gap **710** towards the top side **244** of the second filter frame **216** such that the bottom edge **805** of the filter **217** may no longer constrained or held by the bottom side **245** of the second filter frame **216**. It should be noted that any of the gaps **700**, **702**, **704**, **706**, **710**, **712**, **714**, **716** may provide clearance for the filters **215**, **217** to be removed from the filter frames **214**, **216**, and that the removal of filters is not limited to the method described above. That is, in some embodiments, the filters **215**, **217** may be transitioned into the gaps **702**, **712** towards the bottom sides **241**, **245** of the first and second filter frames **214**, **216** such that the top edges **800**, **804** of the filters **215**, **217** are no longer held or constrained by the top sides **240**, **244** of the first and second filter frames **214**, **216**. In other embodiments, the filters **215**, **217** may be moved into the gaps **704**, **706**, **714**, **716** towards the lateral sides **242**, **243**, **246**, **247** of the first and second filter frames **214**, **216** in a similar fashion such that the filters **215**, **217** are no longer constrained by one or more surfaces of the first and second filter frames **214**, **216**. As noted above, each filter **215**, **217** may include one or more individual filters (e.g., a plurality of filters) that each may be individually removed via movement into one of the gaps as described above.

(68) Providing a collapsible air hood assembly capable of transitioning between a deployed configuration and a collapsed configuration, in accordance with the present disclosure, can reduce the size (e.g., footprint) of the HVAC unit and protect components within the air hood assembly (e.g., filters) during transportation, thereby reducing time and transportation costs associated with the delivery of the HVAC units. Further, by configuring the air hood assembly as described above,

the air hood assembly may provide protection to certain components of the HVAC system that would otherwise need to be removed and reinstalled upon delivery of the HVAC unit, and additional time and costs associated with assembly of the HVAC unit may be reduced.

(69) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(70) While only certain features and embodiments of the disclosure have been illustrated and described, many modifications and changes may occur to those skilled in the art, such as variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, including temperatures and pressures, mounting arrangements, use of materials, colors, orientations, and so forth without materially departing from the novel teachings and advantages of the subject matter recited in the claims. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the disclosure. Furthermore, in an effort to provide a concise description of the exemplary embodiments, all features of an actual implementation may not have been described, such as those unrelated to the presently contemplated best mode of carrying out the disclosure, or those unrelated to enabling the claimed disclosure. It should be noted that in the development of any such actual implementation, as in any engineering or design project, numerous implementation specific decisions may be made. Such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure, without undue experimentation.

Claims

1. A filter assembly for a heating, ventilation, and air conditioning (HVAC) unit, comprising: a filter frame configured to adjustably couple to a housing of the HVAC unit and configured to support a plurality of filters, wherein the filter frame comprises: a first sub-frame configured to support a first filter of the plurality of filters; a second sub-frame configured to support a second filter of the plurality of filters; a gap disposed between the first sub-frame and the second sub-frame; and a flexible material extending across the gap, wherein the flexible material is configured to block air flow between the first sub-frame and the second sub-frame, wherein the filter assembly is configured to adjust between a collapsed configuration and a deployed configuration, and the filter assembly is configured to be contained within the housing in the collapsed configuration.
2. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are rotatably coupled to one another.
3. The filter assembly of claim 2, wherein the first sub-frame and the second sub-frame are rotatably coupled to one another via one or more hinge joints such that the first sub-frame and the second sub-frame rotate toward one another and about the one or more hinge joints during transition of the filter assembly from the deployed configuration to the collapsed configuration.
4. The filter assembly of claim 3, wherein the first sub-frame and the second sub-frame rotate away from one another and about the one or more hinge joints during transition of the filter assembly from the collapsed configuration to the deployed configuration.
5. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are configured to overlap with one another in the collapsed configuration.
6. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are

configured to align with one another in a direction extending from the housing in the deployed configuration.

7. The filter assembly of claim 1, wherein the gap is configured to facilitate transition of the filter assembly from the collapsed configuration to the deployed configuration.

8. The filter assembly of claim 1, wherein the first sub-frame and the second sub-frame are each formed with a respective plurality of C-channels, and wherein each C-channel in the respective plurality of C-channels defines an additional gap configured to receive at least one filter of the plurality of filters.

9. The filter assembly of claim 8, wherein the additional gap is configured to define a clearance between the corresponding C-channel and the at least one filter to enable installation and removal of the at least one filter.

10. A hood assembly for a heating, ventilation, and air conditioning (HVAC) unit, comprising: a top panel configured to rotatably couple to a housing of the HVAC unit, wherein the top panel comprises a panel portion and a lip extending crosswise from the panel portion; and a filter frame configured to rotatably couple to the top panel, wherein the filter frame is configured to support at least one filter, wherein the hood assembly is adjustable between a collapsed configuration and a deployed configuration, the top panel is configured to retain the filter frame within the housing in the collapsed configuration, the lip is configured to engage with a top panel retainer of the housing in the collapsed configuration, and the lip is configured to be spaced apart from the top panel retainer of the housing in the deployed configuration.

11. The hood assembly of claim 10, wherein the filter frame comprises a first sub-frame configured to support a first filter and a second sub-frame configured to support a second filter, and wherein the first sub-frame and the second sub-frame are rotatably coupled to one another.

12. The hood assembly of claim 11, wherein the first sub-frame is configured to rotatably couple to the top panel, and the second sub-frame is configured to rotatably couple to the housing.

13. The hood assembly of claim 10, wherein the hood assembly is configured to transition between the collapsed configuration and the deployed configuration with the at least one filter installed within the filter frame.

14. The hood assembly of claim 10, wherein, in the collapsed configuration, the lip is configured to restrict rotation of the top panel in an inward direction relative to the housing, restrict movement of the top panel in a downward direction, or both, via engagement of the lip with the top panel retainer of the housing.

15. The hood assembly of claim 11, wherein the first sub-frame and the second sub-frame are configured to rotate toward one another during transition of the hood assembly from the deployed configuration to the collapsed configuration, and the first sub-frame and the second sub-frame are configured to rotate away from one another during transition of the hood assembly from the collapsed configuration to the deployed configuration.

16. A heating, ventilation, and air conditioning (HVAC) unit, comprising: a housing comprising a first lip; an intake hood assembly adjustably coupled to the housing and configured to transition between a collapsed configuration and a deployed configuration, wherein the intake hood assembly comprises: a top cover rotatably coupled to the housing, wherein the top cover comprises a second lip, the first lip of the housing and the second lip of the top cover are configured to engage with one another in the collapsed configuration, and the first lip of the housing and the second lip of the top cover are configured to be spaced apart from one another in the deployed configuration; and a filter frame rotatably coupled to the top cover and to the housing, wherein the top cover and the filter frame are configured to rotate relative to one another and relative to the housing during transition of the intake hood assembly between the collapsed configuration and the deployed configuration.

17. The HVAC unit of claim 16, wherein the filter frame comprises a first filter frame configured to support a first filter and a second filter frame configured to support a second filter.

18. The HVAC unit of claim 16, wherein the top cover is configured to overlap with the filter frame

and enclose the filter frame within the housing in the collapsed configuration.

19. The HVAC unit of claim 16, wherein the top cover comprises a panel portion, and the second lip of the top cover extends crosswise from the panel portion.
